Q1.

Human twins may result from:

either the fertilisation of two separate eggs, released together;

or the fertilisation of a single egg which then divides into two cells, each cell developing into a baby.

The table below gives information about three sets of twins.

sets of twins	name	blood group	eye colour	mass in kg	sex
set 1	Sasha Ninvata	A AB	brown brown	37 37	female female
set 2	Lucy Tom	O O	brown brown	38 32	female male
set 3	Fred Jack	O O	blue blue	34 36	male male

(a) Which **one** of the characteristics below **cannot** be used to decide whether twins have come from a single egg or two eggs?
Tick the correct box.

blood group ☐ mass ☐

eye colour ☐ sex ☐

1 mark

(b) (i) Use the information in the table to suggest which set of twins could have come from a single egg.

.....

1 mark

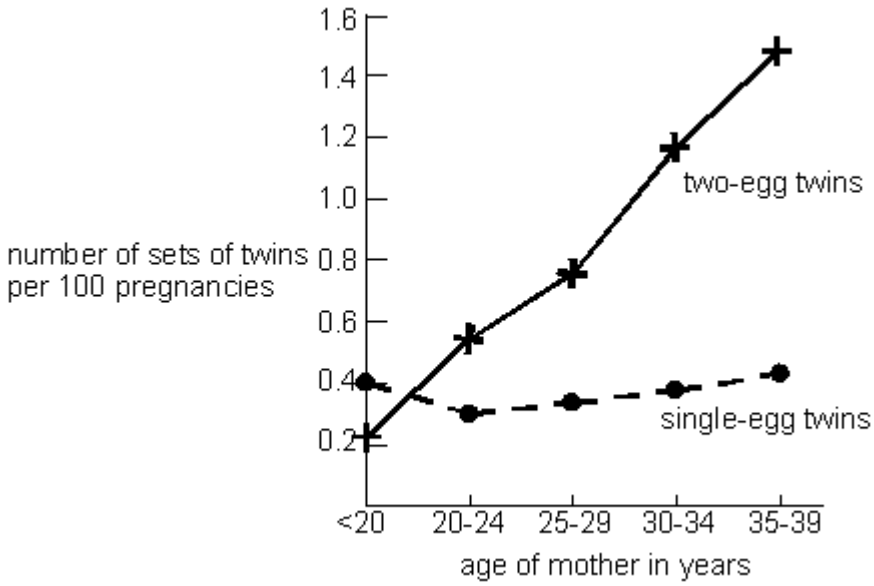
(ii) Explain why you have suggested this set of twins.

.....

.....

1 mark

(c) The graph shows the number of sets of single-egg and two-egg twins born to mothers of different ages.



Two hormones, produced by the pituitary gland, cause eggs to develop in the ovary.

Which statement could explain the results shown on the graph?
Tick the correct box.

As women get older, their ovaries do not respond to either of the hormones.	☐	As women get older, they are less likely to give birth to single-egg twins.	☐
As women get older, they produce more of each hormone.	☐	As women get older, their ovaries release two eggs every month.	☐

1 mark

(d) There are two types of cell division, meiosis and mitosis.

(i) Why are eggs and sperms produced by **meiosis**?

.....

.....

1 mark

(ii) Why does a fertilised egg divide by **mitosis**?

.....

.....

1 mark
Maximum 6 marks

Q1.

(a) mass √

if more than one box is ticked, award no mark

1

(b) (i) set 3 **or** Fred and Jack

consequential marking does not apply accept '3'

1

(ii) any **one** from

- they are similar apart from mass
- they have the same blood group, eye colour and sex
  *consequential marking does **not** apply*
  accept 'they have the same blood group and sex'
  ***both** blood group and sex must be mentioned for the mark*
- all the characteristics determined by genes are the same
  *accept 'because set 1 **or** Sasha and Ninvata have different blood groups and set 2 **or** Lucy and Tom are different sexes'*

1

(c) As woman get older, they produce more of each hormone. ✓

if more than one box is ticked, award no mark

1

(d) (i) any **one** from

- it halves the number of chromosomes
  accept 'it produces 23 chromosomes'
  ***or** 'it halves the genetic material'*
- it maintains the number of chromosomes in the next generation
  accept 'so that the fertilised egg has 46 chromosomes'
- it leads to variation **or** it produces offspring with different characteristics

1

(ii) any **one** from

- so that all the cells have the same genotype
  accept 'so that the genetic information in all the cells is the same'
- so that all the cells have the same number of chromosomes

1

[6]